

Day	outlook	Humidity	Wind	Play & Tennis (arg)
D1	sunny	High	Weak	No
D2	sunny	High	strong	No
D3	overcast	High	Weak	Yes
D4	Rainy	High	Weak	Yes
D5	Rain	Normal	Weak	Yes
D6	Rain	Normal	strong	No

Entropy: $H(S) = -P_{\text{yes}} \times \log_2(P_{\text{yes}}) - P_{\text{no}} \times \log_2(P_{\text{no}})$

$H(S) = 0$: The database is completely pure

$H(S) = 1$: The database has maximum impurity
[Equal split so Yes & No]

Information Gain $IG(S, A)$

Details steps

Step 1: calculate Total Entropy of Dataset $H(S)$

Total NO: of dataset $N=6$

Target = Yes: 3 (D3, D4, D5)

Target = NO: 3 (D1, D2, D6)

Probability

$$P_{\text{yes}} = P_{\text{NO}} = 3/6 = 0.5$$

$$\begin{aligned} H(S) &= 0.5 \times \log_2(0.5) + 0.5 \times \log_2(0.5) \\ &= 0.5 \times (-1) + 0.5 \times (-1) \\ &= 0.5 + 0.5 \\ &= 1.0 \end{aligned}$$

Step 2: calculate information Gain by for each section

Attribute A: outlook.

outlook takes three distinct values: sunny, overcast, and Rain.

→ Sunny (2 instance: D_1, D_2): Target = [NO, NO] [0 Yes, 2 No]
 $H(\text{sunny}) = 0$ (Pure Node)

→ Overcast (1 instance: D_3): Target = [Yes] (1 Yes, 0 No)

→ Rain (3 instance: D_4, D_5, D_6): Target = [Yes, Yes, No]
 [2 Yes, 1 No]

$$P = \frac{2}{3} = 0.667 \quad P = \frac{1}{3} = 0.333$$

$$\begin{aligned} H(\text{Rain}) &= -(0.667) \times \log_2(0.667) - 0.333 \times \log_2(0.333) \\ &= -(0.667) \times (-0.585) - 0.333 \times (-1.585) \\ &= 0.1390 + 0.528 \\ &= 0.667 \end{aligned}$$

Weighted Entropy for outlook

$$E\text{-weighted (outlook)} = \left(\frac{2}{6}\right) \times H(\text{sunny}) + \left(\frac{1}{6}\right) \times H(\text{overcast}) + \left(\frac{3}{6}\right) \times H(\text{Rain})$$

$$\begin{aligned} E\text{-weighted (outlook)} &= \left(\frac{2}{6}\right) \times (0) + \left(\frac{1}{6}\right) \times (0) + \left(\frac{3}{6}\right) \times (0.667) \\ &= 0 + 0 + 0.333 \\ &= 0.333 \end{aligned}$$

$$IG(S, A) = IG(S, \text{outlook})$$

$$= H(S) - W \cdot E(\text{outlook})$$

$$1 - 0.333$$

$$= 0.667$$

$$IG(S, \text{outlook}) = 0.667$$

Attribute B: Humidity

Humidity takes two distinct values: High and Normal.

→ High (4 instances: D_1, D_2, D_3, D_4) : Target = [No, Yes, Yes, Yes] (2 Yes, 2 No)

$$P\text{-Yes} = 3/4 = 0.75, \quad P\text{-No} = 1/4 = 0.25 \Rightarrow H(\text{High}) = 1.0$$

→ Normal (2 instances: D_5, D_6) : Target = [Yes, No] (1 Yes, 1 No)

$$P\text{-Yes} = 1/2 = 0.5, \quad P\text{-No} = 1/2 = 0.5 \Rightarrow H(\text{Normal}) = 1.0$$

→ Weighted Entropy for humidity

$$E\text{-weighted (Humidity)} = \left(\frac{4}{6}\right) * (1.00) + \left(\frac{2}{6}\right) * (1.00) = 1.00$$

$$IG(S, \text{Humidity}) = H(S) - E\text{-weighted (Humidity)} = 1.6 - 1.0 = 0.60$$

Attribute C: wind

wind takes two distinct values: weak and strong

→ strong (4 instances: D_1, D_3, D_4, D_5) : Target = [No, Yes, Yes, Yes]

$$(3, 1, 1, 1) \Rightarrow (3 \text{ Yes}, 1 \text{ No})$$

$$P\text{-Yes} = 3/4 = 0.75; \quad D_2, D_6 : \text{Target} = [\text{No}, \text{No}]$$

$$(0 \text{ Yes}, 2 \text{ No})$$

$$H(\text{strong}) = 0.918$$

Weighted Entropy for wind:

$$E\text{-weighted (wind)} = \left(\frac{4}{6}\right) * (0.918) + \left(\frac{2}{6}\right) * (0) = 0.612$$

$$IG(S, \text{wind}) = 1.6 - 0.612 = 0.988$$

$$= 0.988$$

step 3 : Root Node selection

Attribute	Weighted Entropy	Weighted	Information Gain
outlook	0.459		$0.5 + 1$ (max)
humidity	1.000		0.00
wind	0.541		0.459

Decision : since outlook has the high Information Gain.

step 4 : splitting the Rain subset (Detailed math)

subset S-Rain contains D4, D5 and D6

Day,	Humidity	wind	Play Tennis?
D4	high	weak	Yes
D5	Normal	weak	Yes
D6	Normal	strong	No

Current Entropy of Rain subset : $H(S-Rain) = 0.918$
 Remaining Candidate Attribute : Humidity, wind

A. Candidate Attribute : Humidity (in Rain subset)
 high (1 instance : D4) : Target = [Yes] $\Rightarrow H(High) = 0$

$$P_{Yes} = \frac{1}{2} = 0.5, \quad P_{No} = \frac{1}{2} = 0.5 \Rightarrow H(Normal) = 1.0$$

$$IG(S-Rain - Humidity) = 0.918 - 0.667 = 0.251$$